

1 Scientific justification

We have discovered a new population of massive, likely quiescent galaxies at $3 < z < 5$, a time when the universe was only 1 – 2 Gyr old, using data from our new legacy SHELA Survey which covers an unprecedentedly large volume of 0.25 Gpc^3 at these redshifts ($\sim 20 \text{ deg}^2$). While the photometry of these galaxies prefer a quiescent solution, they are also consistent with dust-obscured heavily star-forming galaxies. We propose an efficient ALMA Band 6 program to observe the redshifted thermal emission from a representative subsample to rule out star-forming solutions at high significance, allowing the use of these distant quiescent galaxies to constrain models of galaxy evolution.

Motivation: Since Edwin Hubble first hand-drew sketches of galaxies observed through an eyepiece almost a century ago, we have known that galaxies in the local universe vary wildly in their star-forming properties, and that the most massive galaxies are primarily quiescent, forming almost no stars, and having an elliptical morphology. One of the key questions in astrophysics is how and when the most massive galaxies become quenched and spheroid-dominated. Theory predicts that in the absence of feedback, massive galaxies should be diskly and star forming. What physical process(es) are responsible for transforming disks into spheroids and quenching star formation? Mergers are a popular candidate, but cosmological simulations have shown that either mergers are not common enough in Λ CDM, or we do not have a good understanding of their effect on galaxies (Porter et al. 2014; Brennan et al. 2015). Violent disk instabilities have been proposed as an alternative or additional mechanism, but the physics remains hotly debated (see discussion in Somerville & Davé 2015). A further question is the interplay between “internal” quenching (perhaps connected to a nuclear black hole) and “environmental” quenching (which includes processes connected to the larger scale density field, such as harassment and tidal and ram pressure stripping).

Constraining Galaxy Evolution with Massive Galaxies at $z > 3$: One can understand the formation of quiescent galaxies by studying the epoch when the first massive galaxies become quenched. While quiescent galaxies can be spectroscopically identified at $z < 2$ (e.g., Dunlop et al. 2007, Cimatti et al. 2004; Bezanson et al. 2018; Estrada-Carpenter et al. 2019), at higher redshifts their faint nature means that they need to be identified via imaging (e.g., Daddi et al. 2005). Interestingly, these galaxies are diverse, as only $\sim 50\%$ of massive ($M > 10^{11} M_\odot$) galaxies at $z \sim 2$ are quiescent, compared to $\sim$ all at $z = 0$ (e.g., Kriek et al. 2006; van Dokkum et al. 2011). This is consistent with simulations, which predict that massive galaxies at high-redshift can be fed through cold-streams, leading to unstable, clumpy disks (e.g., Dekel et al. 2009; Ceverino et al. 2010). Moreover, the black holes that power AGN feedback may not yet have had time to grow in all galaxies.

When and how do the first quiescent galaxies quench? The frontier is now at $z > 3$, where deep near-infrared imaging is crucial to detect light from the less massive stars that dominate the mass. Surveys over small fields find a high ($\sim 40\%$) quiescent fraction amongst massive ($\log[M/M_\odot] > 11$) galaxies at $z = 3.5$ (e.g., Stefanon et al. 2013; Nayyeri et al. 2014; Spitler et al. 2014), while the 1.6 deg^2 UltraVISTA dataset yielded a smaller fraction (16%; Muzzin et al. 2013). Quiescent galaxies have been found to even higher redshifts, as Straatman et al. (2014) found 17 likely quiescent massive galaxies at $z = 4$, and Glazebrook et al. (2017) confirmed a similar object at $z = 3.72$. However, sample and cosmic variance (CV) prevent robust conclusions. The Muzzin et al. sample consisted of only 28 massive quiescent galaxies at $z > 3$, with a sizable CV uncertainty of 17% (Moster et al. 2011). The prevalence of star-formation amongst massive galaxies at $z > 3$ is therefore poorly constrained – we do not know their number density to better than 1 in 5. Do they still represent half of the massive galaxy population (implying that significant quenching must be happening at higher redshift)? Or, as found by Muzzin et al., do star-forming galaxies dominate at these masses?

All of these observations are in tension with theoretical predictions (Figure 1). For example, Brennan et al. (2015) recently compared their massive quiescent galaxy fraction of $\sim 30\%$ at $z = 3$

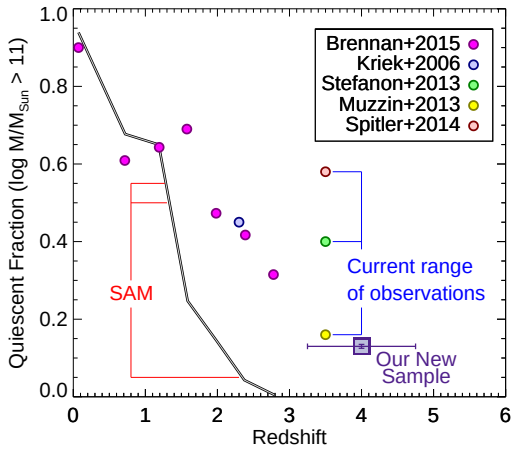


Figure 1: The evolution of the quiescent fraction in massive ($\log M/M_{\odot} > 11$) galaxies. There is significant disagreement in the literature at $z > 3$, by a factor of $> 3\times$. Models predict essentially zero massive quiescent galaxies at $z > 3$, less than *any* of the observations. Our sample (Stevans et al., in prep) of massive quiescent galaxies from our 20 deg² SHELA survey is $> 10\times$ larger than the UltraVISTA sample, allowing us to obtain a more robust measure of the quiescent fraction at $z = 3-4$ (dark purple point). This ALMA program will prove the quiescent nature of a representative cross-section of this sample.

from CANDELS to a semi-analytic model, where passive spheroids are built up via major and minor mergers and disk instabilities. These simulations predict a quiescent fraction which is in agreement with observations at $z \leq 1$, but drops to zero by $z \sim 2.5$, in significant tension with the observations discussed above. The Illustris simulations have found that some massive quiescent $z \sim 2$ galaxies in their models formed the bulk of their stars at higher redshifts, implying that their quiescent fraction at $z > 3$ is non-zero, though likely still small (Wellons et al. 2015). Clearly, models are missing physical processes at high redshift which lead to a higher fraction of massive galaxies being quenched. If the quiescent fraction at $z = 3-4$ is truly as high as at $z = 1-2$, it will necessitate significant revision of the physical processes in these models.

Breaking New Ground with SHELA: To further constrain models of the changing physical processes regulating star formation at high redshift, much larger samples of massive, quiescent (and star-forming) galaxies are needed. To make progress, we have undertaken a 20 deg² multi-wavelength survey: the Spitzer-HETDEX Exploratory Large Area Survey (SHELA), covering an area $> 10\times$ larger than UltraVISTA. This entire field is covered by CTIO/DECam optical ($m_{5\sigma, AB} \sim 25$; PI Papovich; Wold et al. 2019), KPNO/NEWMFIRM near-infrared ($m_{5\sigma, AB} \sim 23$; PI Finkelstein; Stevans et al., in prep), and Spitzer/IRAC mid-infrared ($m_{5\sigma, AB} \sim 22.5$; Papovich et al. 2016) imaging, probing an unprecedented volume of 0.25 Gpc³ at $z = 3-4$. Over this volume, our estimated uncertainty due to cosmic variance for $\log(M/M_{\odot}) > 11$ galaxies is only 4%; $4\times$ less than that from UltraVISTA. Our survey provides the *key* combination of depth and area necessary to make significant progress on isolating the onset of the first quenched galaxies and bulges.

We have constructed a multi-wavelength (0.4–4.5 μm) *K*-band-selected catalog over this full region, using the Tractor (Lang et al. 2016) to deblend the IRAC photometry, and EAZY (Brammer et al. 2008) to measure photometric redshifts and stellar population properties for all 1.53 million catalogued galaxies (Stevans et al., in prep). We have selected a parent sample of 3684 massive $z \sim 3-5$ galaxies as those with a significant detection in *K* and both IRAC bands, $> 60\%$ of their integrated photometric redshift probability density at $3 < z < 5$, and $\log(M_*/M_{\odot}) > 11$ (consistent with the number density of massive galaxies at $z \sim 4$ from Deshmukh et al. 2018 in UltraVISTA). Of these massive galaxies, 506 are likely quiescent, with $\log \text{sSFR} < -11 \text{ yr}^{-1}$. The implied quiescent fraction is thus $13.7 \pm 0.7\%$ (including both Poisson and CV uncertainties), close to the value from UltraVISTA, though at much higher significance (and most notably, a non-zero value, inconsistent with predictions from simulations). We show three example objects from our sample in Figure 2.

However, not included in the error budget is the potential for sample contamination. The main contaminants are dusty star-forming galaxies, either at similar redshifts, or at $z \sim 1$ (where the dusty Balmer break masquerades as the Lyman break). Before we can use our results to constrain the physical processes in state-of-the-art simulations, we must first validate the quiescent nature of our sample. Here we propose to do that with an efficient ALMA program.

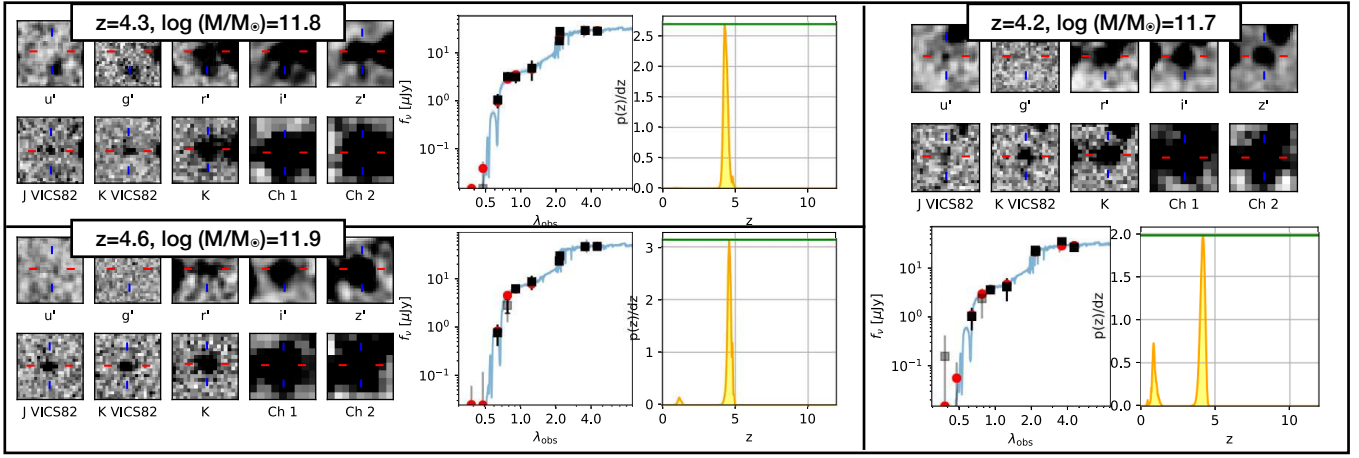


Figure 2: Candidate quiescent objects in our sample. Each panel shows 7'' stamps in all imaging bands, the observed SED (black), the best-fit EAZY model (blue line; red circles), and the EAZY photo- z probability distribution. These sources are characterized by little significant optical emission, significant 2–5 μm emission, and a flat K –[4.5] color, consistent with a quiescent galaxy at $z \sim 3$ –5. ALMA 1.1mm observations will rule out dusty star-forming solutions at both low and high redshift.

2 Description of observations

Our immediate observing goal is to reject the possibility that galaxies in our sample of candidate massive quiescent high-redshift galaxies are in reality massive dust-obscured star-forming galaxies. Both types of galaxies appear red in the observed optical-to-mid-IR. However, true passive galaxies should have little-to-no dust emission in the far-IR, while dusty SFGs will be producing most of their luminous emission in that range. Far-IR constraints are thus crucial to the fidelity of our quiescent sample. This is illustrated in Figure 3, which shows our best-fitting EAZY model (using the updated EAZY-py software) to the photometry of a typical galaxy in our sample at $z = 4.15$, compared to the best-fitting maximal star-forming model from MAGPHYS (da Cunha et al. 2008), as well as

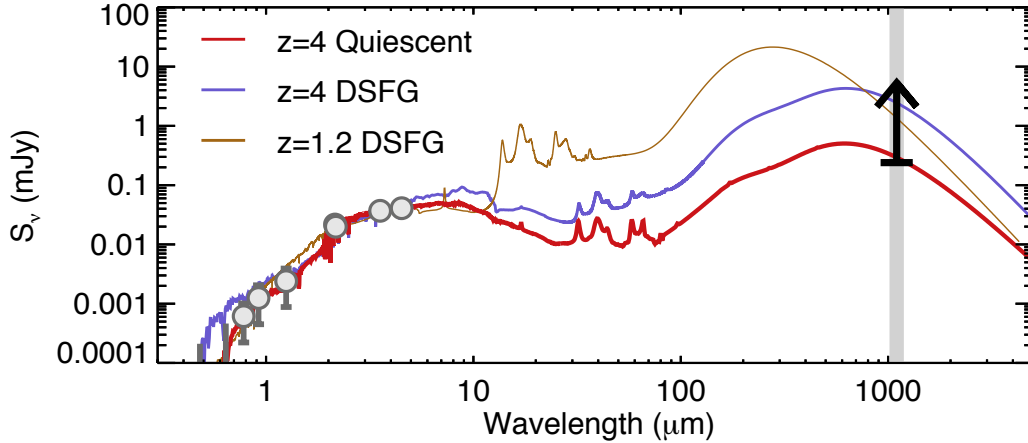


Figure 3: The photometry of a typical object in our sample ($K \sim 20.5$; gray circles), which is best-fit at $z = 4.15$ galaxy with $\log(M_*/M_\odot) = 11.9$, and a $\log \text{sSFR} = -11.5 \text{ yr}^{-1}$, indicating it is a massive galaxy which has quenched at a very early epoch. The full SED of this best-fitting population is shown in red. Although the data prefer the quiescent high-redshift solution, it is also consistent with a $z \sim 4$ galaxy with significant star-formation ($\text{SFR} \sim 250 M_\odot \text{ yr}^{-1}$; $\log \text{sSFR} = -9.5$; in purple), and a lower redshift dusty star-forming galaxy (DSFG) at $z \sim 1.2$ ($\text{SFR} \sim 30 M_\odot \text{ yr}^{-1}$; $\log \text{sSFR} = -9.7$; brown). Our proposed 3σ continuum depth of 240 μJy is shown by the black arrow. This depth significantly rules out star-forming galaxies at any redshift (our primary science goal), and is capable of setting secure limits on low-levels of obscured star-formation in mostly quiescent galaxies.

Table 1: *Properties of our observations and target sample.*

Frequency (GHz)	Sensitivity (mJy)	N_{galaxies}	Redshift	K-band mag (AB)	Stellar Mass ($M_{\odot}$)	sSFR (yr^{-1})
265	0.08	100	3.5–5.0	20–22	11.1–11.9	–11 to –11.6

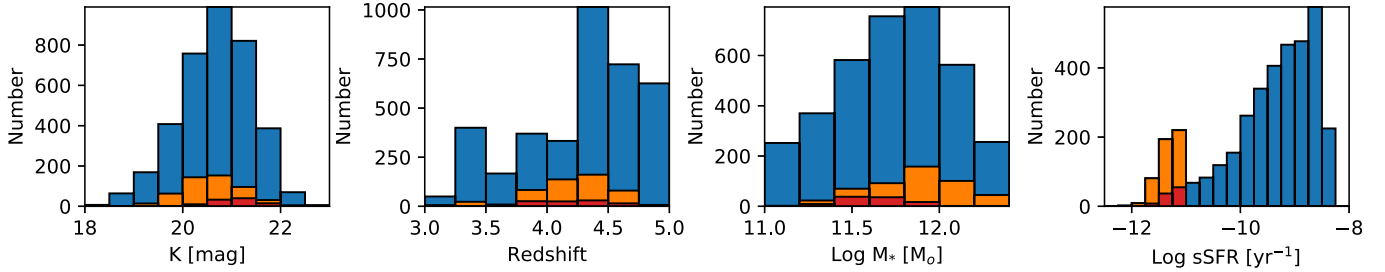


Figure 4: The distribution of the SHELA massive $z=3\text{--}5$ galaxy sample (blue), the quiescent sub-sample (orange), and those selected for observations (red). This figure highlights that the sources we propose to observe span the full sample range of redshift, stellar mass, sSFR and K-band magnitude.

the best-fitting MAGPHYS model at the secondary EAZY redshift of $z = 1.2$. As shown in Figure 3, rest-frame far-IR (1.1 mm) observations to $\text{rms}=80 \mu\text{Jy}$ with ALMA can robustly rule out the star-forming solutions in minutes per source (see technical justification for details on sensitivity and frequency choices). While the SHELA field has been observed by *Herschel*/SPIRE at 0.25–0.5mm (PI Viero; Viero et al. 2014), these data are very shallow, sensitive to obscured SFRs $> 200 M_{\odot} \text{ yr}^{-1}$ at $z \sim 1$, and $> 1000 M_{\odot} \text{ yr}^{-1}$ at $z \sim 3$, much higher than the expected SFRs of the contaminants (Figure 3). Nonetheless, we have compared our quiescent sample to the catalog from Viero et al. (2014), and found no robust matches.

Proposed Sample: We propose a large pilot study to observe 100 of our candidate quiescent galaxies, with our target redshifts, stellar masses, and sSFRs drawn randomly to sample the dynamic range of the parent sample, as shown by the red histograms in Figure 4. Even though this is only $\sim 20\%$ of our parent sample of 506 objects, it is a factor of $> 6\times$ larger than the previous largest sample of quiescent $z \sim 4$ galaxies from Straatman et al. (2014), and their result was based on stacking from shallow *Herschel* data. With these observations we can efficiently estimate the contamination fraction in our sample at reasonable confidence, in just a few hours of ALMA time.

We draw our target sample to cover $3.5 < z < 5$, $11.1 < \log(M/M_{\odot}) < 11.9$, and $20 < K < 22$, comparable to the parent sample. We can reach our target rms over these 100 objects in 7.7 hours of ALMA Band 6 time (inclusive of overheads), accomplishing our primary science goal.

3 References

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